

# Irene Hou

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Cognitive Science Department

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Google Scholar 

## Research Interests

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Human-Centered AI, Human-Computer Interaction, participatory AI, computing pedagogy

## Education

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| 2024 – Present<br><i>La Jolla, CA</i> | <b>University of California, San Diego</b><br>Ph.D. in Cognitive Science — Advisor: Philip Guo<br><i>Anticipated graduation: 06/2029</i>       |
| 2018 – 2022<br><i>La Jolla, CA</i>    | <b>University of California, San Diego</b><br>B.S. in Cognitive Science (Design/Interaction Spec.), Computer Science minor<br><i>cum laude</i> |

## Fellowships and Awards

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| Apr 2026 | <b>Paul and Daisy Soros Fellowship for New Americans</b><br>Awarded nationally to 30 of 3,000+ applicants in 2026 (1% selection rate)                       |
| Apr 2025 | <b>National Science Foundation Graduate Research Fellowship (NSF GRFP)</b><br>Awarded nationally to 1,000 of 15,000+ applicants in 2025 (6% selection rate) |
| Jan 2023 | <b>CES 2023 Innovation Honoree</b><br>UX Lead · FluentPet Connect                                                                                           |
| Jan 2015 | <b>Bronze Medalist</b> , National eLit Illuminary Awards<br>Author · <i>Snowspirit: The Virgo Key</i>                                                       |

## Publications

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| CHI 2027<br>(under review) | [10] <b>Irene Hou</b> , Xiaotian Su, and April Wang. Grading Builds Tacit Knowledge: How Generative AI Grading Risks Estranging Instructors and TAs from Their Students. <i>In ACM CHI Conference on Human Factors in Computing Systems, ACM CHI '27, 2027.</i>                              |
| CHI 2027<br>(under review) | [9] <b>Irene Hou</b> <sup>*</sup> , Jeongeon Park <sup>*</sup> , Risa Sundu <sup>*</sup> , and Steven Dow. DreamPlace: An AI-Powered Elicitation Platform to Help Citizens Envision Community Spaces. <i>In ACM CHI Conference on Human Factors in Computing Systems, ACM CHI '27, 2027.</i> |
| CHI 2027<br>(under review) | [8] Kate Hamilton, <b>Irene Hou</b> , Dev Patel, Sheena Nnam, Hena Patel, and Stephen MacNeil. “Stuck in a Spiral”: Understanding Shame and Guilt in Computing Students’ AI Use. <i>In ACM CHI Conference on Human Factors in Computing Systems, ACM CHI '27, 2027.</i>                      |
| CHI 2026                   | [7] <b>Irene Hou</b> , Alex Qin, Lauren Cheng, and Philip Guo. <b>Beyond the Desk: Barriers and Future Opportunities for AI to Assist Scientists in Embodied Physical Tasks.</b> <i>In ACM CHI Conference on Human Factors in Computing Systems, ACM CHI '26, 2026.</i>                      |
| CHI 2026                   | [6] <b>Irene Hou</b> , Zeyu Xiong, Philip Guo, and April Wang. <b>“Bespoke Bots”: Diverse Instructor Needs for Customizing Generative AI Classroom Chatbots.</b> <i>In ACM CHI Conference on Human Factors in Computing Systems, ACM CHI '26, 2026.</i>                                      |

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| CI 2025     | [5] <b>Irene Hou*</b> , Jeongeon Park*, Risa Sundu*, and Steven Dow. <b>Value-Centered Framing Supports Inclusive Idea Convergence in Participatory Civic Design</b> . In <i>ACM Collective Intelligence, ACM CI '25</i> , 2025.                                                                                                                                    |
| ITiCSE 2025 | [4] <b>Irene Hou</b> , Kate Hamilton, Owen Man, Srishty Muthusekaran, Jeffin Johnykutty, Leili Zadeh, and Stephen MacNeil. <b>“All Roads Lead to ChatGPT”: How Generative AI is Eroding Social Interactions and Student Learning Communities</b> . In <i>Innovation and Technology in Computer Science Education, ACM ITiCSE '25</i> , 2025.                        |
| ACE 2025    | [3] Sebastian Gutierrez, <b>Irene Hou</b> , Jihye Lee, Kenneth Angelikas, Owen Man, Sophia Mettelle, James Prather, Paul Denny, and Stephen MacNeil. <b>Seeing the Forest and the Trees: Solving Visual Graph and Tree Based Data Structure Problems using Large Multimodal Models</b> . In <i>Australasian Computing Education Conference, ACM ACE '25</i> , 2025. |
| ACE 2024    | [2] <b>Irene Hou</b> , Sophia Mettelle, Owen Man, Zhuo Li, Cynthia Zastudil, and Stephen MacNeil. <b>The Effects of Generative AI on Introductory Students’ Help Seeking Preferences</b> . In <i>Australasian Computing Education Conference, ACM ACE '24</i> , 2024.                                                                                               |
| ACE 2024    | [1] <b>Irene Hou</b> , Owen Man, Sophia Mettelle, Sebastian Gutierrez, Kenneth Angelikas, and Stephen MacNeil. <b>More Robots are Coming: Large Multimodal Models (ChatGPT) can Solve Visually Diverse Images of Parsons Problems</b> . In <i>Australasian Computing Education Conference, ACM ACE '24</i> , 2024.                                                  |

(\* denotes co-first author)

## Posters, Extended Abstracts, Workshop Papers

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| SIGCSE 2025 | [p.1] <b>Irene Hou</b> , Hannah V. Nguyen, Owen Man, and Stephen MacNeil. <b>The Evolving Usage of Generative AI by Computing Students</b> . In <i>Technical Symposium on Computing Science Education, ACM SIGCSE '25</i> , 2025. Poster. |
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## Research Experience

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| Summer 2026<br>Zürich, CH        | <b>PEACH Lab, ETH Zürich</b> · Visiting Researcher · <i>Advised by April Wang</i><br>· AI-assisted grading and instructor tacit knowledge [10]                                 |
| 2024 – Present<br>La Jolla, USA  | <b>Design Lab, UC San Diego</b> · Graduate Researcher · <i>Advised by Philip Guo</i><br>· AI for scientists’ embodied lab and field work [7]                                   |
| 2024 – Present<br>La Jolla, USA  | <b>ProtoLab, UC San Diego</b> · Graduate Researcher · <i>Advised by Steven Dow</i><br>· Participatory civic design and AI-powered community elicitation [5, 9]                 |
| Summer 2025<br>Zürich, CH        | <b>PEACH Lab, ETH Zürich</b> · Visiting Researcher · <i>Advised by April Wang</i><br>· Instructor needs for customizing classroom AI chatbots [6]                              |
| 2023 – 2025<br>Philadelphia, USA | <b>Temple HCI Lab, Temple University</b> · Research Lead · <i>Advised by Stephen MacNeil</i><br>· AI impact on computing students; led 5 undergraduate research teams [1–4, 8] |
| 2021 – 2022<br>La Jolla, USA     | <b>ProtoLab, UC San Diego</b> · Undergraduate Researcher<br>· Designed for CoBoards, a digital whiteboard system                                                               |

Spring 2021  
*La Jolla, USA*      **Comparative Cognition Lab, UC San Diego** · Undergraduate Research Assistant  
· Designed interface for a video-analysis behavioral encoding tool

## Additional Work Experience

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2021 – 2023  
*San Jose, USA*      **FluentPet** · UX Lead  
· Led designers and developers in the end-to-end design, build, and launch of FluentPet Connect, connected-hardware app and product (CES 2023 Innovation Honoree)  
· Launch achieved 72% DAU and 78.7% WAU, with 58k+ weekly interactions in first month

Summer 2020  
*San Diego, USA*      **SONY** · Curriculum Designer  
· Designed curriculum for SONY's KOOV (edTech robotics), used by 10+ instructional teams

Summer 2017  
*San Diego, USA*      **Booknection Creative Writing Summer Camp** · Founder  
· Founded a creative science-fiction writing camp for teen writers  
· Designed original curriculum, pitched to local academies, secured funding, instructed classes

2013 – 2018  
*San Diego, USA*      **Snowspirit: The Virgo Key** · Author  
· Published an award-winning science-fiction/fantasy novel  
· Fundraised, designed PR campaigns, organized launch events with 2,000+ attendees

## Invited Talks and Media

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Apr 2026      **Design@Large, UC San Diego Design Lab** · Invited Discussant  
Session: Learning in the Age of GenAI: Practice, Policy, and Pedagogy

Apr 2026      **Center for Teaching, Learning and Outreach**, California Institute of Technology  
Talk: How GenAI is Eroding Social Interactions and Student Learning Communities

Jan 2026      **The CS-Ed Podcast** · Guest  
Episode S4xE14: [GenAI's Impact on Student Help Seeking and More](#)

Jun 2025      **Responsible Tech Institute Student Network**, University of Oxford  
Talk: How GenAI is Eroding Social Interactions and Student Learning Communities

May 2024      **Raspberry Pi Foundation**, United Kingdom (remote)  
Seminar: [Generative AI is changing undergraduate education and undergraduate research](#)

## Teaching

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2021 – 2025  
*La Jolla, CA*      **Teaching Assistant** · UC San Diego, Department of Cognitive Science  
COGS 124: HCI Technical Systems Research (Spring 2025)  
COGS 127: Data-Driven UX/Product Design (Winter 2025, Winter 2021)  
COGS 13: Field Methods: Studying Cognition in the Wild (Winter 2022)

2017 – 2022  
*San Diego, CA*      **Instructor** · K–12 Education  
Grace Academy (2019–2022): Robotics, Python, Physics, Computational Thinking  
Starton Tech EDU (2018–2020): Python, Computational Thinking, KOOV Robotics  
All-Star Academy (2017–2019): Creative Writing, Introduction to Programming

## Guest Lectures

May 2026      COGS 10: Cognitive Consequences of Technology, UC San Diego  
Lecture: Social Impacts of AI in Education

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|----------|---------------------------------------------------------------------------------------------------------------------------------|
| May 2025 | Canyon Crest Academy, San Diego<br>Seminar: Introduction to HCI and Exploring HCI Careers                                       |
| Feb 2025 | COGS 124: HCI Technical Systems Research, UC San Diego<br>Lecture: Fairness, Accountability, Trust, Ethics in Human-Centered AI |

## Academic Service

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| Reviewer          | CHI ('24, '25, '26), DIS '26, SIGCSE '26 |
| Student Volunteer | SIGCSE '25                               |

## Leadership and Outreach

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| 2025 – Present | <b>Vice President, Graduate Women in Computing @ UC San Diego</b><br>Organized and launched the Undergraduate Research Night series (3 events, 2025–2026) |
| Winter 2025    | <b>Outreach Co-Chair, Graduate Women in Computing @ UC San Diego</b>                                                                                      |

## Research Mentorship

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Undergraduates I have directly supervised or mentored in research. ( <sup>†</sup> denotes community college student )

**Owen Man<sup>†</sup>**, De Anza College [1–4, p.1] → MS, CSU San Marcos → Microsoft

**Kenneth Angelikas**, Temple University [2–3] → Microsoft

**Kate Hamilton**, Temple University [4, 8] → Comcast

**Ayah Ibrahim**, UC San Diego → PhD, UC Santa Cruz

**Olivia Lam<sup>†</sup>**, Palomar College → UCLA (Data Science)

**Brandson Trinh<sup>†</sup>**, De Anza College → UC San Diego (Data Science)

**Zoe Kathari**, UC San Diego (Honors Thesis, Cognitive Science)

**Alex Qin**, UC San Diego [7]

**Lauren Cheng**, UC San Diego [7]

**Sophia Mettelle**, Temple University [1–2]

**Sebastian Gutierrez**, Temple University [2–3]

**Zhuo Li**, Temple University [1]

**Srishty Muthusekaran**, Temple University [4]

**Jeffin Johnykutty**, Temple University [4]

**Leili Massoum Zadeh**, Temple University [4]

**Hannah Vy Nguyen**, Temple University [p.1]

**Ben Nguyen**, Temple University

**Alexander Yu**, UC San Diego